

## Vector Differentiation

**THE VECTOR DIFFERENTIAL OPERATOR DEL**, written  $\nabla$ , is defined by

$$\nabla \equiv \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \equiv \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

This vector operator possesses properties analogous to those of ordinary vectors. It is useful in defining three quantities which arise in practical applications and are known as the *gradient*, the *divergence* and the *curl*. The operator  $\nabla$  is also known as *nabla*.

**THE GRADIENT.** Let  $\phi(x, y, z)$  be defined and differentiable at each point  $(x, y, z)$  in a certain region of space (i.e.  $\phi$  defines a differentiable scalar field). Then the *gradient* of  $\phi$ , written  $\nabla\phi$  or  $\text{grad } \phi$ , is defined by

$$\nabla\phi = \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \phi = \frac{\partial\phi}{\partial x} \mathbf{i} + \frac{\partial\phi}{\partial y} \mathbf{j} + \frac{\partial\phi}{\partial z} \mathbf{k}$$

Note that  $\nabla\phi$  defines a vector field.

The component of  $\nabla\phi$  in the direction of a unit vector  $\mathbf{a}$  is given by  $\nabla\phi \cdot \mathbf{a}$  and is called the directional derivative of  $\phi$  in the direction  $\mathbf{a}$ . Physically, this is the rate of change of  $\phi$  at  $(x, y, z)$  in the direction  $\mathbf{a}$ .

**THE DIVERGENCE.** Let  $\mathbf{V}(x, y, z) = V_1 \mathbf{i} + V_2 \mathbf{j} + V_3 \mathbf{k}$  be defined and differentiable at each point  $(x, y, z)$  in a certain region of space (i.e.  $\mathbf{V}$  defines a differentiable vector field). Then the *divergence* of  $\mathbf{V}$ , written  $\nabla \cdot \mathbf{V}$  or  $\text{div } \mathbf{V}$ , is defined by

$$\begin{aligned} \nabla \cdot \mathbf{V} &= \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \cdot (V_1 \mathbf{i} + V_2 \mathbf{j} + V_3 \mathbf{k}) \\ &= \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z} \end{aligned}$$

Note the analogy with  $\mathbf{A} \cdot \mathbf{B} = A_1 B_1 + A_2 B_2 + A_3 B_3$ . Also note that  $\nabla \cdot \mathbf{V} \neq \mathbf{V} \cdot \nabla$ .

**THE CURL.** If  $\mathbf{V}(x, y, z)$  is a differentiable vector field then the *curl* or *rotation* of  $\mathbf{V}$ , written  $\nabla \times \mathbf{V}$ ,  $\text{curl } \mathbf{V}$  or  $\text{rot } \mathbf{V}$ , is defined by

$$\begin{aligned}\nabla \times \mathbf{V} &= \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \times (V_1 \mathbf{i} + V_2 \mathbf{j} + V_3 \mathbf{k}) \\&= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_1 & V_2 & V_3 \end{vmatrix} \\&= \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_2 & V_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ V_1 & V_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ V_1 & V_2 \end{vmatrix} \mathbf{k} \\&= \left( \frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z} \right) \mathbf{i} + \left( \frac{\partial V_1}{\partial z} - \frac{\partial V_3}{\partial x} \right) \mathbf{j} + \left( \frac{\partial V_2}{\partial x} - \frac{\partial V_1}{\partial y} \right) \mathbf{k}\end{aligned}$$

Note that in the expansion of the determinant the operators  $\frac{\partial}{\partial x}$ ,  $\frac{\partial}{\partial y}$ ,  $\frac{\partial}{\partial z}$  must *precede*  $V_1, V_2, V_3$ .

## SOLVED PROBLEMS

### THE GRADIENT

1. If  $\phi(x, y, z) = 3x^2y - y^3z^2$ , find  $\nabla\phi$  (or  $\text{grad } \phi$ ) at the point  $(1, -2, -1)$ .

$$\begin{aligned}\nabla\phi &= \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) (3x^2y - y^3z^2) \\&= \mathbf{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \mathbf{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) + \mathbf{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2) \\&= 6xy \mathbf{i} + (3x^2 - 3y^2z^2) \mathbf{j} - 2y^3z \mathbf{k} \\&= 6(1)(-2) \mathbf{i} + \{3(1)^2 - 3(-2)^2(-1)^2\} \mathbf{j} - 2(-2)^3(-1) \mathbf{k} \\&= -12 \mathbf{i} - 9 \mathbf{j} - 16 \mathbf{k}\end{aligned}$$

3. Find  $\nabla\phi$  if (a)  $\phi = \ln |\mathbf{r}|$ , (b)  $\phi = \frac{1}{r}$ .

(a)  $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ . Then  $|\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}$  and  $\phi = \ln |\mathbf{r}| = \frac{1}{2} \ln(x^2 + y^2 + z^2)$ .

$$\begin{aligned}\nabla\phi &= \frac{1}{2}\nabla\ln(x^2 + y^2 + z^2) \\ &= \frac{1}{2}\left\{\mathbf{i}\frac{\partial}{\partial x}\ln(x^2 + y^2 + z^2) + \mathbf{j}\frac{\partial}{\partial y}\ln(x^2 + y^2 + z^2) + \mathbf{k}\frac{\partial}{\partial z}\ln(x^2 + y^2 + z^2)\right\} \\ &= \frac{1}{2}\left\{\mathbf{i}\frac{2x}{x^2 + y^2 + z^2} + \mathbf{j}\frac{2y}{x^2 + y^2 + z^2} + \mathbf{k}\frac{2z}{x^2 + y^2 + z^2}\right\} = \frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{x^2 + y^2 + z^2} = \frac{\mathbf{r}}{r^2}\end{aligned}$$

$$\begin{aligned}(b) \nabla\phi &= \nabla\left(\frac{1}{r}\right) = \nabla\left(\frac{1}{\sqrt{x^2 + y^2 + z^2}}\right) = \nabla\{(x^2 + y^2 + z^2)^{-1/2}\} \\ &= \mathbf{i}\frac{\partial}{\partial x}(x^2 + y^2 + z^2)^{-1/2} + \mathbf{j}\frac{\partial}{\partial y}(x^2 + y^2 + z^2)^{-1/2} + \mathbf{k}\frac{\partial}{\partial z}(x^2 + y^2 + z^2)^{-1/2} \\ &= \mathbf{i}\left\{-\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}2x\right\} + \mathbf{j}\left\{-\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}2y\right\} + \mathbf{k}\left\{-\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}2z\right\} \\ &= \frac{-x\mathbf{i} - y\mathbf{j} - z\mathbf{k}}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{\mathbf{r}}{r^3}\end{aligned}$$

4. Show that  $\nabla r^n = nr^{n-2}\mathbf{r}$ .

$$\begin{aligned}\nabla r^n &= \nabla(\sqrt{x^2 + y^2 + z^2})^n = \nabla(x^2 + y^2 + z^2)^{n/2} \\ &= \mathbf{i}\frac{\partial}{\partial x}\{(x^2 + y^2 + z^2)^{n/2}\} + \mathbf{j}\frac{\partial}{\partial y}\{(x^2 + y^2 + z^2)^{n/2}\} + \mathbf{k}\frac{\partial}{\partial z}\{(x^2 + y^2 + z^2)^{n/2}\} \\ &= \mathbf{i}\left\{\frac{n}{2}(x^2 + y^2 + z^2)^{n/2-1}2x\right\} + \mathbf{j}\left\{\frac{n}{2}(x^2 + y^2 + z^2)^{n/2-1}2y\right\} + \mathbf{k}\left\{\frac{n}{2}(x^2 + y^2 + z^2)^{n/2-1}2z\right\} \\ &= n(x^2 + y^2 + z^2)^{n/2-1}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) \\ &= n(r^2)^{n/2-1}\mathbf{r} = nr^{n-2}\mathbf{r}\end{aligned}$$

Note that if  $\mathbf{r} = r\mathbf{r}_1$  where  $\mathbf{r}_1$  is a unit vector in the direction  $\mathbf{r}$ , then  $\nabla r^n = nr^{n-1}\mathbf{r}_1$ .

5. Show that  $\nabla\phi$  is a vector perpendicular to the surface  $\phi(x,y,z) = c$  where  $c$  is a constant.

Let  $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$  be the position vector to any point  $P(x,y,z)$  on the surface. Then  $d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$  lies in the tangent plane to the surface at  $P$ .

$$\text{But } d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = 0 \quad \text{or} \quad \left(\frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}\right) \cdot (dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}) = 0$$

i.e.  $\nabla\phi \cdot d\mathbf{r} = 0$  so that  $\nabla\phi$  is perpendicular to  $d\mathbf{r}$  and therefore to the surface.

6. Find a unit normal to the surface  $x^2y + 2xz = 4$  at the point  $(2, -2, 3)$ .

$$\nabla(x^2y + 2xz) = (2xy + 2z)\mathbf{i} + x^2\mathbf{j} + 2x\mathbf{k} = -2\mathbf{i} + 4\mathbf{j} + 4\mathbf{k} \quad \text{at the point } (2, -2, 3).$$

$$\text{Then a unit normal to the surface} = \frac{-2\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}}{\sqrt{(-2)^2 + (4)^2 + (4)^2}} = -\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}.$$

Another unit normal is  $\frac{1}{3}\mathbf{i} - \frac{2}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$  having direction opposite to that above.

7. Find an equation for the tangent plane to the surface  $2xz^2 - 3xy - 4x = 7$  at the point  $(1, -1, 2)$ .

$$\nabla(2xz^2 - 3xy - 4x) = (2z^2 - 3y - 4)\mathbf{i} - 3x\mathbf{j} + 4xz\mathbf{k}$$

Then a normal to the surface at the point  $(1, -1, 2)$  is  $7\mathbf{i} - 3\mathbf{j} + 8\mathbf{k}$ .

The equation of a plane passing through a point whose position vector is  $\mathbf{r}_0$  and which is perpendicular to the normal  $\mathbf{N}$  is  $(\mathbf{r} - \mathbf{r}_0) \cdot \mathbf{N} = 0$ . (See Chap. 2, Prob. 18.) Then the required equation is

$$[(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) - (\mathbf{i} - \mathbf{j} + 2\mathbf{k})] \cdot (7\mathbf{i} - 3\mathbf{j} + 8\mathbf{k}) = 0$$

or

$$7(x-1) - 3(y+1) + 8(z-2) = 0.$$

10. Find the directional derivative of  $\phi = x^2yz + 4xz^2$  at  $(1, -2, -1)$  in the direction  $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$ .

$$\begin{aligned} \nabla\phi &= \nabla(x^2yz + 4xz^2) = (2xyz + 4z^2)\mathbf{i} + x^2\mathbf{j} + (x^2y + 8xz)\mathbf{k} \\ &= 8\mathbf{i} - \mathbf{j} - 10\mathbf{k} \quad \text{at } (1, -2, -1). \end{aligned}$$

The unit vector in the direction of  $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$  is

$$\mathbf{a} = \frac{2\mathbf{i} - \mathbf{j} - 2\mathbf{k}}{\sqrt{(2)^2 + (-1)^2 + (-2)^2}} = \frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$$

Then the required directional derivative is

$$\nabla\phi \cdot \mathbf{a} = (8\mathbf{i} - \mathbf{j} - 10\mathbf{k}) \cdot \left(\frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}\right) = \frac{16}{3} + \frac{1}{3} + \frac{20}{3} = \frac{37}{3}$$

Since this is positive,  $\phi$  is increasing in this direction.

11. (a) In what direction from the point  $(2, 1, -1)$  is the directional derivative of  $\phi = x^2yz^3$  a maximum?  
 (b) What is the magnitude of this maximum?

$$\begin{aligned}\nabla\phi &= \nabla(x^2yz^3) = 2xyz^3\mathbf{i} + x^2z^3\mathbf{j} + 3x^2yz^2\mathbf{k} \\ &= -4\mathbf{i} - 4\mathbf{j} + 12\mathbf{k} \quad \text{at } (2, 1, -1).\end{aligned}$$

Then by Problem 9,

- (a) the directional derivative is a maximum in the direction  $\nabla\phi = -4\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}$ ,  
 (b) the magnitude of this maximum is  $|\nabla\phi| = \sqrt{(-4)^2 + (-4)^2 + (12)^2} = \sqrt{176} = 4\sqrt{11}$ .

12. Find the angle between the surfaces  $x^2 + y^2 + z^2 = 9$  and  $z = x^2 + y^2 - 3$  at the point  $(2, -1, 2)$ .

The angle between the surfaces at the point is the angle between the normals to the surfaces at the point.

A normal to  $x^2 + y^2 + z^2 = 9$  at  $(2, -1, 2)$  is

$$\nabla\phi_1 = \nabla(x^2 + y^2 + z^2) = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k} = 4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$$

A normal to  $z = x^2 + y^2 - 3$  or  $x^2 + y^2 - z = 3$  at  $(2, -1, 2)$  is

$$\nabla\phi_2 = \nabla(x^2 + y^2 - z) = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k} = 4\mathbf{i} - 2\mathbf{j} - \mathbf{k}$$

$(\nabla\phi_1) \cdot (\nabla\phi_2) = |\nabla\phi_1| |\nabla\phi_2| \cos \theta$ , where  $\theta$  is the required angle. Then

$$\begin{aligned}(4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}) \cdot (4\mathbf{i} - 2\mathbf{j} - \mathbf{k}) &= |4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}| |4\mathbf{i} - 2\mathbf{j} - \mathbf{k}| \cos \theta \\ 16 + 4 - 4 &= \sqrt{(4)^2 + (-2)^2 + (4)^2} \sqrt{(4)^2 + (-2)^2 + (-1)^2} \cos \theta\end{aligned}$$

$$\text{and } \cos \theta = \frac{16}{6\sqrt{21}} = \frac{8\sqrt{21}}{63} = 0.5819; \text{ thus the acute angle is } \theta = \arccos 0.5819 = 54^\circ 25'.$$

## THE DIVERGENCE

15. If  $\mathbf{A} = x^2z \mathbf{i} - 2y^3z^2 \mathbf{j} + xy^2z \mathbf{k}$ , find  $\nabla \cdot \mathbf{A}$  (or  $\text{div } \mathbf{A}$ ) at the point  $(1, -1, 1)$ .

$$\begin{aligned}\nabla \cdot \mathbf{A} &= \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \cdot (x^2z \mathbf{i} - 2y^3z^2 \mathbf{j} + xy^2z \mathbf{k}) \\ &= \frac{\partial}{\partial x}(x^2z) + \frac{\partial}{\partial y}(-2y^3z^2) + \frac{\partial}{\partial z}(xy^2z) \\ &= 2xz - 6y^2z^2 + xy^2 = 2(1)(1) - 6(-1)^2(1)^2 + (1)(-1)^2 = -3 \quad \text{at } (1, -1, 1).\end{aligned}$$

16. Given  $\phi = 2x^3y^2z^4$ . (a) Find  $\nabla \cdot \nabla \phi$  (or  $\text{div grad } \phi$ ).

(b) Show that  $\nabla \cdot \nabla \phi = \nabla^2 \phi$ , where  $\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$  denotes the Laplacian operator

$$\begin{aligned}(a) \nabla \phi &= \mathbf{i} \frac{\partial}{\partial x}(2x^3y^2z^4) + \mathbf{j} \frac{\partial}{\partial y}(2x^3y^2z^4) + \mathbf{k} \frac{\partial}{\partial z}(2x^3y^2z^4) \\ &= 6x^2y^2z^4 \mathbf{i} + 4x^3yz^4 \mathbf{j} + 8x^3y^2z^3 \mathbf{k}\end{aligned}$$

$$\begin{aligned}\text{Then } \nabla \cdot \nabla \phi &= \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \cdot (6x^2y^2z^4 \mathbf{i} + 4x^3yz^4 \mathbf{j} + 8x^3y^2z^3 \mathbf{k}) \\ &= \frac{\partial}{\partial x}(6x^2y^2z^4) + \frac{\partial}{\partial y}(4x^3yz^4) + \frac{\partial}{\partial z}(8x^3y^2z^3) \\ &= 12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2\end{aligned}$$

$$\begin{aligned}(b) \nabla \cdot \nabla \phi &= \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \cdot \left( \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} \right) \\ &= \frac{\partial}{\partial x} \left( \frac{\partial \phi}{\partial x} \right) + \frac{\partial}{\partial y} \left( \frac{\partial \phi}{\partial y} \right) + \frac{\partial}{\partial z} \left( \frac{\partial \phi}{\partial z} \right) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \\ &= \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi = \nabla^2 \phi\end{aligned}$$

17. Prove that  $\nabla^2 \left( \frac{1}{r} \right) = 0$ .

$$\nabla^2 \left( \frac{1}{r} \right) = \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \left( \frac{1}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$\frac{\partial}{\partial x} \left( \frac{1}{\sqrt{x^2+y^2+z^2}} \right) = \frac{\partial}{\partial x} (x^2+y^2+z^2)^{-1/2} = -x(x^2+y^2+z^2)^{-3/2}$$

$$\begin{aligned} \frac{\partial^2}{\partial x^2} \left( \frac{1}{\sqrt{x^2+y^2+z^2}} \right) &= \frac{\partial}{\partial x} [-x(x^2+y^2+z^2)^{-3/2}] \\ &= 3x^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2} = \frac{2x^2-y^2-z^2}{(x^2+y^2+z^2)^{5/2}} \end{aligned}$$

Similarly,

$$\frac{\partial^2}{\partial y^2} \left( \frac{1}{\sqrt{x^2+y^2+z^2}} \right) = \frac{2y^2-z^2-x^2}{(x^2+y^2+z^2)^{5/2}} \quad \text{and} \quad \frac{\partial^2}{\partial z^2} \left( \frac{1}{\sqrt{x^2+y^2+z^2}} \right) = \frac{2z^2-x^2-y^2}{(x^2+y^2+z^2)^{5/2}}$$

$$\text{Then by addition, } \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \left( \frac{1}{\sqrt{x^2+y^2+z^2}} \right) = 0.$$

The equation  $\nabla^2 \phi = 0$  is called *Laplace's equation*. It follows that  $\phi = 1/r$  is a solution of this equation.

## THE CURL

23. If  $\mathbf{A} = xz^3 \mathbf{i} - 2x^2yz \mathbf{j} + 2yz^4 \mathbf{k}$ , find  $\nabla \times \mathbf{A}$  (or curl  $\mathbf{A}$ ) at the point  $(1, -1, 1)$ .

$$\begin{aligned} \nabla \times \mathbf{A} &= \left( \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \times (xz^3 \mathbf{i} - 2x^2yz \mathbf{j} + 2yz^4 \mathbf{k}) \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix} \\ &= \left[ \frac{\partial}{\partial y} (2yz^4) - \frac{\partial}{\partial z} (-2x^2yz) \right] \mathbf{i} + \left[ \frac{\partial}{\partial z} (xz^3) - \frac{\partial}{\partial x} (2yz^4) \right] \mathbf{j} + \left[ \frac{\partial}{\partial x} (-2x^2yz) - \frac{\partial}{\partial y} (xz^3) \right] \mathbf{k} \\ &= (2z^4 + 2x^2y) \mathbf{i} + 3xz^2 \mathbf{j} - 4xyz \mathbf{k} = 3\mathbf{j} + 4\mathbf{k} \quad \text{at } (1, -1, 1). \end{aligned}$$

24. If  $\mathbf{A} = x^2y \mathbf{i} - 2xz \mathbf{j} + 2yz \mathbf{k}$ , find curl curl  $\mathbf{A}$ .

$$\begin{aligned} \text{curl curl } \mathbf{A} &= \nabla \times (\nabla \times \mathbf{A}) \\ &= \nabla \times \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & -2xz & 2yz \end{vmatrix} = \nabla \times [(2x+2z)\mathbf{i} - (x^2+2z)\mathbf{k}] \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x+2z & 0 & -x^2-2z \end{vmatrix} = (2x+2)\mathbf{j} \end{aligned}$$

27. Prove: (a)  $\nabla \times (\nabla \phi) = \mathbf{0}$  (curl grad  $\phi = \mathbf{0}$ ), (b)  $\nabla \cdot (\nabla \times \mathbf{A}) = 0$  (div curl  $\mathbf{A} = 0$ ).

$$\begin{aligned}
 (a) \quad \nabla \times (\nabla \phi) &= \nabla \times \left( \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k} \right) \\
 &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix} \\
 &= \left[ \frac{\partial}{\partial y} \left( \frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left( \frac{\partial \phi}{\partial y} \right) \right] \mathbf{i} + \left[ \frac{\partial}{\partial z} \left( \frac{\partial \phi}{\partial x} \right) - \frac{\partial}{\partial x} \left( \frac{\partial \phi}{\partial z} \right) \right] \mathbf{j} + \left[ \frac{\partial}{\partial x} \left( \frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left( \frac{\partial \phi}{\partial x} \right) \right] \mathbf{k} \\
 &= \left( \frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right) \mathbf{i} + \left( \frac{\partial^2 \phi}{\partial z \partial x} - \frac{\partial^2 \phi}{\partial x \partial z} \right) \mathbf{j} + \left( \frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right) \mathbf{k} = \mathbf{0}
 \end{aligned}$$

provided we assume that  $\phi$  has continuous second partial derivatives so that the order of differentiation is immaterial.

$$\begin{aligned}
 (b) \quad \nabla \cdot (\nabla \times \mathbf{A}) &= \nabla \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\
 &= \nabla \cdot \left[ \left( \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \mathbf{i} + \left( \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \mathbf{j} + \left( \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \mathbf{k} \right] \\
 &= \frac{\partial}{\partial x} \left( \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) + \frac{\partial}{\partial y} \left( \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) + \frac{\partial}{\partial z} \left( \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \\
 &= \frac{\partial^2 A_3}{\partial x \partial y} - \frac{\partial^2 A_2}{\partial x \partial z} + \frac{\partial^2 A_1}{\partial y \partial z} - \frac{\partial^2 A_3}{\partial y \partial x} + \frac{\partial^2 A_2}{\partial z \partial x} - \frac{\partial^2 A_1}{\partial z \partial y} = 0
 \end{aligned}$$

assuming that  $\mathbf{A}$  has continuous second partial derivatives.

Note the similarity between the above results and the results  $(\mathbf{C} \times \mathbf{C}m) = (\mathbf{C} \times \mathbf{C})m = \mathbf{0}$ , where  $m$  is a scalar and  $\mathbf{C} \cdot (\mathbf{C} \times \mathbf{A}) = (\mathbf{C} \times \mathbf{C}) \cdot \mathbf{A} = 0$ .

29. Prove  $\nabla \times (\nabla \times \mathbf{A}) = -\nabla^2 \mathbf{A} + \nabla(\nabla \cdot \mathbf{A})$ .

$$\begin{aligned}
 \nabla \times (\nabla \times \mathbf{A}) &= \nabla \times \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\
 &= \nabla \times \left[ \left( \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \mathbf{i} + \left( \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \mathbf{j} + \left( \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \mathbf{k} \right] \\
 &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} & \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} & \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \end{vmatrix} \\
 &= \left[ \frac{\partial}{\partial y} \left( \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) - \frac{\partial}{\partial z} \left( \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \right] \mathbf{i} \\
 &\quad + \left[ \frac{\partial}{\partial z} \left( \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) - \frac{\partial}{\partial x} \left( \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \right] \mathbf{j} \\
 &\quad + \left[ \frac{\partial}{\partial x} \left( \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) - \frac{\partial}{\partial y} \left( \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \right] \mathbf{k} \\
 &= \left( -\frac{\partial^2 A_1}{\partial y^2} - \frac{\partial^2 A_1}{\partial z^2} \right) \mathbf{i} + \left( -\frac{\partial^2 A_2}{\partial z^2} - \frac{\partial^2 A_2}{\partial x^2} \right) \mathbf{j} + \left( -\frac{\partial^2 A_3}{\partial x^2} - \frac{\partial^2 A_3}{\partial y^2} \right) \mathbf{k} \\
 &\quad + \left( \frac{\partial^2 A_2}{\partial y \partial x} + \frac{\partial^2 A_3}{\partial z \partial x} \right) \mathbf{i} + \left( \frac{\partial^2 A_3}{\partial z \partial y} + \frac{\partial^2 A_1}{\partial x \partial y} \right) \mathbf{j} + \left( \frac{\partial^2 A_1}{\partial x \partial z} + \frac{\partial^2 A_2}{\partial y \partial z} \right) \mathbf{k} \\
 &= \left( -\frac{\partial^2 A_1}{\partial x^2} - \frac{\partial^2 A_1}{\partial y^2} - \frac{\partial^2 A_1}{\partial z^2} \right) \mathbf{i} + \left( -\frac{\partial^2 A_2}{\partial x^2} - \frac{\partial^2 A_2}{\partial y^2} - \frac{\partial^2 A_2}{\partial z^2} \right) \mathbf{j} + \left( -\frac{\partial^2 A_3}{\partial x^2} - \frac{\partial^2 A_3}{\partial y^2} - \frac{\partial^2 A_3}{\partial z^2} \right) \mathbf{k} \\
 &\quad + \left( \frac{\partial^2 A_1}{\partial x^2} + \frac{\partial^2 A_2}{\partial y \partial x} + \frac{\partial^2 A_3}{\partial z \partial x} \right) \mathbf{i} + \left( \frac{\partial^2 A_1}{\partial x \partial y} + \frac{\partial^2 A_2}{\partial y^2} + \frac{\partial^2 A_3}{\partial z \partial y} \right) \mathbf{j} + \left( \frac{\partial^2 A_1}{\partial x \partial z} + \frac{\partial^2 A_2}{\partial y \partial z} + \frac{\partial^2 A_3}{\partial z^2} \right) \mathbf{k} \\
 &= -\left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (A_1 \mathbf{i} + A_2 \mathbf{j} + A_3 \mathbf{k}) \\
 &\quad + \mathbf{i} \left( \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) + \mathbf{j} \left( \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) + \mathbf{k} \left( \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) \\
 &= -\nabla^2 \mathbf{A} + \nabla \left( \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) \\
 &= -\nabla^2 \mathbf{A} + \nabla(\nabla \cdot \mathbf{A})
 \end{aligned}$$

## Some Definitions:

- A vector field with zero divergence is said to be **solenoidal**.
- A vector field with zero curl is said to be **irrotational**.
- A scalar field with zero gradient is said to be, er, well, **constant**.

## Laplacian

The second-order differential operator inside the  $()$  on the RHS is called the *Laplace operator* or simply the *Laplacian*; it is usually denoted by the triangle  $\Delta$ ,

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

Algebraically, the Laplacian is the scalar square of the  $\nabla$  vector,

$$\begin{aligned}\Delta &= \nabla^2 = \nabla \cdot \nabla = \frac{\partial}{\partial x} \frac{\partial}{\partial x} + \frac{\partial}{\partial y} \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \frac{\partial}{\partial z} \\ &= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\end{aligned}$$

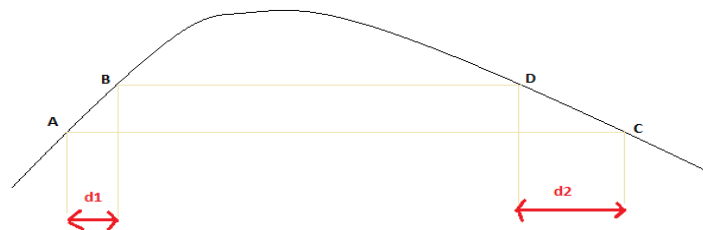
## Q. What is the physical significance of divergence, curl and gradient?

### GRADIENT:

Gradient is the rate of change of a parameter P with respect to parameter Q

For Example,

1. In the image below, the surface from point A to B is much steeper when compared to the surface from point C to D. That is, the magnitude of slope AB is greater than the magnitude of slope CD (Slope is the rate of change of height with respect to distance). Hence, the gradient of surface AB is greater than the gradient of surface CD.

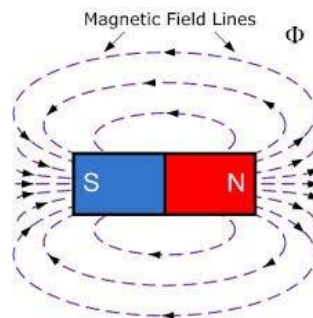


## DIVERGENCE:

Divergence is the net flow of a field/liquid/substance out of a unit volume.

For Example,

1. In a pipe, If, Net flow of liquid into the pipe = Net flow of liquid out of the pipe, then the Divergence=0.
2. Consider a Magnet. The Magnetic field lines originate at one pole and end at the other pole. Thus, Divergence is 0. In fact, this is Gauss's law for Magnetism.



## CURL:

Curl gives the measure of the angular velocity of an object.

If Curl is zero, it means the object is not rotating.

If Curl is not zero, its magnitude represents the speed of the object, and its direction denotes the axis of rotation.